Humidity Box

Element A

Problem Statement

An art teacher teaching at Eleanor Roosevelt High School, located in Greenbelt, Maryland, is facing the problem of maintaining the dry state of paper and clay for her students' art projects, who are all in grades 9-12. The root of the problem is the humidity that has been ever-present within her classroom, which has been an issue since the art teacher started teaching at Eleanor Roosevelt High School about 2 years ago. The humidity in the room is a prominent issue because it interferes with the students' work, curriculum progress, and learning. Due to the humidity, the air in the room is heavy, and paper-based artwork is increasingly difficult to work with. Clay takes longer to dry and sometimes loses shape due to the heat that it collects. Over time, humid air can cause mold & insects, which can lead to health and safety problems within the classroom. Mold can cause staining and illness, while insects can chew on the paper and create holes in it, leaving the artwork damaged and likely with eggs in it. The papers can be affected by the fluctuating humidity, leading to the dimensions shrinking or being rendered unusable.

Background Research

1.) Humidity of 70% or higher adjacent to a surface can cause serious damage to the paper and artwork. The Health and Safety Executive recommends that relative humidity indoors should be maintained at 40-70%, while other experts recommend that the range should be 30-60%. Most people find 30-60% most comfortable. (1)

2.) Smaller indoor spaces tend to have more moisture or air quality problems. This is because they are not as exposed to sun, wind, or natural ventilation that would help to eliminate humidity. Dehumidifiers would help with the issue. (2)

3.) During the winter months, it is advisable to keep your indoor humidity at the lower end of the aforementioned 40-60% range. In fact, in some homes, it is often a good idea to drop below 40% relative humidity to protect your home and maintain indoor comfort levels. The colder winter air is dry, and although a home may be well sealed, the change to winter conditions will cause the humidity in said home to drop naturally. If a home is not sealed well, the cold air can infiltrate the indoor spaces and cause the humidity to drop significantly. (6)

4.) When you add a plastic sheet, people should raise the inner temperature above the dew point. If moving the condensation to somewhere else less noticeable. Water vapor always moves from warm to cold and normally condensation takes place on the nearest cold surface - usually a window. If condensation does not form on a window, it may be forming somewhere else less noticeable, or the indoor temperature may be high enough to hold that amount of water vapor. (4)

5.) For many people, a dew point below 13°C (55°F) feels dry, but beginning above 18°C (65°F) often feels muggy. The highest dew point ever recorded was 35°C (95°F) in Saudi Arabia. In comparison, areas of the United States that are very humid, like the southern state of Mississippi and Washington, D.C., often have dew points of around 27°C (80°F). Areas close to warm bodies of water typically have higher humidity because warm water can evaporate into water vapor more easily than cold water, which adds more water into the air. (3)

6.) There are respiratory issues with humidity, airborne-transmitted infectious bacteria and viruses,of allergenic mite and fungal populations. (5)

Research References

Bannister, Marie. How Humidity Damages Your Home — and How to Fight It. www.airthings.com/resources/home-humidity-damage#. (Bannister)

Managing (And understanding the importance of) indoor humidity levels. (n.d.). https://www.napleshomes.com/indoor-humidity-guide.php

Conditioning, A. U. P. a. T. C. H. a. A. (2019, November 18). High indoor humidity problems in winter - around the clock. Around the Clock. https://aroundclock.com/blog/high-indoor-humidity-problems-in-winter/

High winter humidity - fine homebuilding. (2013, October 29). Fine Homebuilding. https://www.finehomebuilding.com/forum/high-winter-humidity

All about humidity. (n.d.). https://education.nationalgeographic.org/resource/all-about-humidity/

Arundel, A V., et al. "Indirect Health Effects of Relative Humidity in Indoor Environments." Environmental Health Perspectives, vol. 65, 1986, pp. 351-361, https://doi.org/10.1289/ehp.8665351. Accessed 7 Mar. 2024.

Current Situation

A traditional art classroom at Eleanor Roosevelt High School is facing a humidity issue which damages paper and clay, rendering it unusable for curriculum based assignments which may utilize those mediums. The dampness prevents paper from drying and creates an environment in which mold can fester and grow, and bugs can nestle into the damp materials, breeding or causing an infestation. The goal is to create a solution that finds a way to keep the unused paper and clay dry, while helping artwork that needs to dry do so quicker.

Problem Justification

This project is important because it will positively impact the art classrooms' environment and aid students by providing a place where they can access art materials, which in turn, allows them to utilize other creative outlets and mediums, previously rendered unattainable due to the humidity. This should be an important cause for the unaffected to rally for because the same aspects of the environmental factors limiting potential outcomes can be applied to real world issues. The primary stakeholders of said potential outcomes are the art teacher and her students because the success of the design will directly impact them, and improve future access to physical mediums. The secondary stakeholders, are the student engineers, and the teacher sponsor because the administration at Eleanor Roosevelt High School will have to approve of certain changes to the art room and the costs of the project. The student engineers will be the ones working on the project, possibly impacting them directly if they are future students of the art teachers', and the teacher sponsor who will be supervising the project.